Portable Coordinate Measurement Arm Joint Design

Project Proposal

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2026

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Declaration

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Institutional Acknowledgement

This declaration aligns with Stellenbosch University's Position Statement on the ethical use of AI in Research and Teaching-Learning-Assessment[1], the Draft interim SU guidelines on allowable AI use and academic integrity in assessment[2], and draws on international best practices.

[1] Position Statement on Ethical use of Artificial Intelligence in Research and Teaching-Learning Assessment

[2] Draft interim SU guidelines on allowable AI use and academic integrity in assessment

Executive Summary

Title of Project
Portable Coordinate Measurement Arm Joint Design
Objectives
Include force analysis due to the implementation of a position-holding mechanism on the joint. To design, build, and test the joint design to be modular. To evaluate the effectiveness and efficiency of the joint design.
What is current practice and what are its limitations?
Current joints are passive and have little to no holding torque. These joints rely on the operator to grip and hold the arm in position during measurement.
What is new in this project?
The proposed joint design implements position-holding mechanism designed from quantitative specifications derived from an error propagation model.
If the project is successful, how will it make a difference?
Implementation of the joint design would minimise measurement error during calibration and operation. A modified kinematic model of the PCMA would be derived to create a more complete framework for PCMA accuracy analysis. The modularity of the joint design would mean it could be implemented across multiple PCMA models.
What are the risks to the project being a success? Why is it expected to be successful?
Manufacturing tolerances might not meet the derived requirements. The position-holding mechanism might introduce unwanted friction that impedes the free movement of the PCMA. The project is expected to succeed because the design requirements that based off a an established error model.
What contributions have/will other students made/make?
Leon Hall, a Stellenbosch Alumna developed the error propagation model that will determine the joint-tolerance derivation for the project.
Which aspects of the project will carry on after completion and why?
The modified error model incorporating force analysis and joint-testing methodology, are intended to form the basis of future PCMA development. The project could also be pursued further to develop a full arm assembly.
What arrangements have been/will be made to expedite continuation?
Extensive documentation of the project will allow future students or researchers to build onto this work without redoing the groundwork.

Table of Contents

List of Figures

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List of Abbreviations

3D three-dimensional

AACMM Articulated Arm Coordinate Measuring Machine

CAD Computer-Aided Design

CMM Coordinate Measurement Machine

D-H Denavit-Hartenberg

FEA Finite-Element Analysis

FIA first article inspection

PCMA Portable Coordinate Measurement Arm

SPAT Single Point Articulation Test

TPM technical performance measures

1 Introduction

1.1 Background

A Portable Coordinate Measurement Arm (PCMA) is a handheld metrology instrument that captures precise three-dimensional (3D) coordinates directly from a physical object [1]. A probe is used to take measurements of an object that are plotted on a 3D plane as coordinates to create a working model [1].

Figure 1: A Portable Coordinate Measurement Arm

PCMAs can also referred to as a Articulated Arm Coordinate Measuring Machine (AACMM)[3] in literature, offer flexible metrology for multiple applications, such as dimensional analysis, Computer-Aided Design (CAD) inspection, reverse engineering etc [3]. The main limitation of these joints is the "human factor", in which operators often introduce fluctuating contact forces, positional instability, and unpredictable probing trajectories that can cause dynamic structural deformations of the arm [4]. It is also important to note that the calibration procedures require the arm to be held completely still [5].

As a result, designing an effective joint with a position-holding mechanism is an important engineering requirement and a good practice of safety design. By designing the joint in this manner, the measurements will become ~~independent of~~ human error, which creates kinematic stability, improving the repeatability and accuracy of the arm.

1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on the joint of a portable measurement coordinate arm without jeopardising its accuracy and ease of movement.

[3]Articulated refers to a mechanical structure composed of a series of rigid linkages connected by rotary joints [2]

The objectives of the project ~~include the following points~~:

1. To go beyond the conventional Denavit-Hartenberg (D-H) model by including force analysis due to the implementation of a position-holding mechanism on the joint.

2. To design, build, and test the joint design to be modular and applicable to a range of PMCAs in industry.

3. To evaluate the effectiveness and efficiency of the joint design by verifying accuracy of measurements, through Single Point Articulation Test (SPAT).

1.3 Scope

in reference to the highlighted, I think it was redundant of me to say as I was trying to limit modularity away from specialised PCMAs but PCMAs are specialised in and of themselves

The consideration of PCMAs will be reduced to simulation, in that data sheets, user and reference manuals will be used to obtain information to model each respective PCMA for their D-H models. Only three of the most widely used PCMAs in industry will be considered to influence modularity of the joint design. Only one joint will be designed, built, tested for data acquisition, through which calibration and accuracy testing will be conducted utilising the modified D-H (which includes force analysis) [5].

1.4 Motivation

PCMAs are becoming more and more preferred over traditional CMMs for in-line[4] and on-site industrial measurement [1]. This is mainly due to their portability, large working volume relative to their footprint, and their cost-effectiveness [3]. They have broad applicability across multiple industries such as aerospace, automotive, heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.

Despite the PCMAs growing popularity, the revolute joints of these arms present an engineering challenge that has received limited investigation in literature. The joints must satisfy competing demands simultaneously:

- Tight run-out tolerances for kinematic accuracy
- Sufficient holding torque to keep the arm stationary once released
- Internal cable routing
- Compact, lightweight form that does not degrade overtime

This.has proved difficult since there are no commercially available joints that meet these demands.

It has been established that angular position errors at each joint propagate through

[4] In the context of PCMAs, "in-line" refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory [1]

the kinematic chain directly affecting end-effector accuracy [6]. This ~~proves~~ that ~~improvements to established~~ joint designs have a measurable impact on overall arm performance [6]. This project addresses that gap by deriving joint specifications from a modified error model based of an error model that was developed by a former Stellenbosch University student, Leon Hall [7]. This will produce a validated joint design and testing methodology to support future PCMA development at Stellenbosch University.

1.5 Use of AI Tools

Gemini is an AI tool that will be used for to check grammatical errors and sentence structure. Claude AI will be used to assist in the formatting of the report on the online platform Overleaf using LaTeX.

1.6 Report Overview

This report follows the following structure:

- Section 1 introduces the project background, aim, objectives, scope, and motivation.
- Section 2 presents a literature review discussing both industry-facing and academic sources relevant to PCMA joint design, kinematic modelling, and error analysis.
- Section 3 describes the engineering research design methodology to be followed.
- Section 4 outlines the workplan for the project.
- Section 5 details the resource requirements and estimated costs.
- Section 6 describes the expected results of the project.
- Section 7 presents the project schedule as a Gantt chart.
- Section 8 provides a risk assessment of the project.
- Section 9 concludes the proposal for the project.

2 Literature Review

PCMAs have emerged as a versatile and important tool in industrial metrology [1]. Compared to the traditional fixed Coordinate Measurement Machine (CMM), the PCMA offers portability and larger measurement volumes [1]. CMMs were original created for inspecting the geometric paths of bent tube, allowing operators to capture 3D coordinates at the point of manufacture [3]. This ~~is the~~ eliminated cost and delays associated with transporting component to a dedicated metrology laboratory [3]. Despite the practical advantages, the measurement accuracy of PCMAs remains a constant challenge caused by interrelating error sources.

The following reviews a few contributions to this field. First, the practical benefits and operational principles of PCMAs are put into context through industry-facing sources from Leussink Engineering and Oasis Alignment Services. The review will then discuss the academic literature covering Cuesta et al.'s virtual simulation study of dynamic deformations, Ibaraki and Saito's novel kinematic model with angular position measurement errors, Santolaria et al.'s calibration procedure based on laser tracker multilateration and lastly Brau and others' design and validation of a self-driven joint model. It must be noted that the literature reviwed for this project will be updated regularly as the project progresses.

2.1 Industry-Facing Contributions

Th~~is~~ following papers give context and explain the contributions of each source to the project from industry.

2.1.1 The Advantages of Using a Portable CMM Arm in Metrology

~~This article discusses the use of PCMAs, in particular, articulated measuring arms for accurate 3D measurement of physical objects in multiple environments such as workshops and production settings [1]. The article goes on to highlight the methodology of capturing precise measurements directly from the object to be measured [1].~~ ~~This allows~~ for real-time tracking and comparison to CAD data without moving any components. The data is transferred to CAD software, which generates 3D models and measurement reports [1]. This facilitates quality control and inspection processes. The importance of understanding the specific use of measurement data is also emphasised [1]. ~~This can also advise interested parties when~~ selecting the appropriate CMM arm and software platform.

2.1.2 Measurement Tool Spotlight: Portable Measuring Arms

This article discusses the evolution of PCMAs, which were originally designed for inspecting bent tubes [3]. This highlights their long-standing presence in the study of metrology. The article emphasises the advantages of PCMAs as opposed to traditional CMMs, specifically in terms of portability, cost-effectiveness, and efficiency in many applications [3]. It is also stated that PCMAs are excellent tools in reverse engineering, CAD-based inspection, and first article inspection (FIA) [5] [3].

2.2 Academic-Facing Contriutions

This following papers give context and explain the contributions of each source to the project from academic studies.

[5] An FIA is a formal, comprehensive verification process that ensures a manufacturer's initial production run meets all engineering drawings, specifications, and design requirements before mass production [8]

2.2.1 Dynamic Deformations in Coordinate Measuring Arms Virtual Simulation

This research paper discuss the errors induced by the structure of CMMs [4]. This highlights the significant impact of deflection caused by contact forces that occur during operation, which is noted as a key area of analysis in prior work [4]. The paper references the work of Vrhovec and Munih, who studied the deflection of CMM segments, discussing their method, which was noted to be expensive and challenging to implement [4]. There are existing standards such as VDI/VDE 2617 and ISO 10360-12 that focus on verification and calibration, however, they lack guidelines for improving accuracy and controlling probing forces [4]. It is important to note that this suggests a gap in literature. The paper takes note of the fact previous studies have shown that measurement errors can reach micrometers due to unaccounted contact forces and deflection [4]. It also addresses ~~an occurrence regarding~~ the reliability of measurements due to human factors and the complexity of the CMM structure, leading to ~~the~~ non-uniform contact forces and unpredictable probing trajectories caused by operators [4].

2.2.2 Novel Kinematic Model of Articulated Arm Coordinate Measuring Machine with Angular Position Measurement Errors of Rotary Axes

This research paper indicates that measurement errors in rotary axis angular positions are mainly due to gradient errors of rotary encoder scales, which vary with angular position [6]. Previous studies have extensively used the D-H model, which includes position and orientation errors of rotary axes as error sources [6]. There is also an acknowledged challenge in ensuring sphere displacement of the stylus [6] constrained sufficiently during SPAT due to the stiffness of the stylus holder [6]. It was noted that there exists a gap in existing models regarding the angular position measurement errors of rotary axes, which implies the need for a new or modified kinematic model. The proposed modification to the SPAT procedure aim to improve the accuracy of angular position measurements [6]. This address the limitations found in previous methodologies and a ~~kind of~~ framework that can be used to derive a model specific to the purposes of this project.

2.2.3 Articulate Arm Coordinate Measuring Machine Calibration by Laser Tracking Multilateration

This research paper discusses different evaluation methods for measuring arms based on established standards such as UNE-EN ISO 10360, ASME B89.4.1, and VDI/VDE 2617 [9]. This highlights their relevance in current practices in industrial metrology. The paper takes note of a gap in research, which is focused on the calibration of AACMMs [9]. This indicates that there is a lack of standardized verification and calibration procedures [9]. The paper also reveals that previous works have aimed

[6] A stylus in a PCMA is a precision probe tip that makes physical contact with an object to capture measurement points [1]

There is no difference. Different terminology for the same thing.

to assess the performance of measuring arms, with some proposing methods to estimate measurement uncertainty and error sources [9]. A comparison of results with ANSI/ASME B89 volumetric performance tests is referenced, which shows good agreement and cost reduction [9]. This ultimately implies a concurrency on the effectiveness of certain evaluation methods [9]. This emphasizes the importance of multilateration[7]. techniques in improving measurement accuracy and reducing uncertainty and it reflects ongoing discussions in the metrology field regarding optimal measurement strategies [9].

2.2.4 Design and Validation of Self-Driven Joint Model for Articulation Arm Coordinate Measuring Machines

This research paper recognises that PCMAs known for their portability and cost-effectiveness, however, it is noted that they face challenges related to measurement accuracy as a result of structural parameter errors and user-induced variability [5]. Key technologies in PCMAs include kinematics modelling, error analysis, and parameter calibration, with various methods proposed for improving accuracy [5]. These methods include non-linear least squares and genetic algorithms [5]. The paper highlights the importance of temperature error compensation and the influence of external forces on measurement accuracy [5]. This indicates that there is a consensus on the need for robust error correction models [5]. Human influences, including inconsistent contact forces and measurement techniques, are identified as significant sources of error [5]. This leads to discussions on the reliability and repeatability of manual measurements [5]. The paper proposes a novel self-driven joint model to enhance measurement automation and accuracy, which addresses the limitations of traditional manual operation in PCMAs [5].

2.2.5 Error Distribution of an Articulated Arm Measurement System

This research paper discusses the lack of a universal standard for calibrating articulated arms, with the ASME standard not being widely adopted [7]. This leads ~~to~~ users to rely on OEM claims regarding accuracy [7]. Previous studies, such as studies by Conrad et al. and Santolaria et al., indicate that errors in articulated arms can average 0.3 mm [7], however, Renaud et al. achieved an accuracy below 1 mm using a vision-based system [7]. When these results are contrasted with the limitations of laser-based systems, which only provide positional data, it highlights a need for thorough investigations into accuracy [7]. The paper suggests that measurement errors are often not constant or linear throughout the measurement volume, which challenges OEM assertions of accuracy [7]. The paper emphasizes the importance of understanding error distribution to better plan the application of articulated arms, as errors can vary significantly across the measurement volume [7].

[7] In the field of metrology, multilateration is a high-precision calibration and measurement technique that determines the 3D position of a probe or target by measuring its distance from multiple, separated, known fixed locations, normally using laser trackers [9]

2.2.6 Shigley's Mechanical Engineering Design, Tenth Edition

The textbook provides different areas of relevance to the joint design of PCMAs, where stiffness, deflection control, and repeatability are the important engineering requirements [10]. Chapter 2 discusses material selection methodology, which includes the specific stiffness (stiffness-to-weight) index[10]. This supports the choice of lightweight, high-modulus materials such as aluminium alloys or carbon fibre composites for PCMAs segments [10]. This decreases gravitational deflection at extended reach positions and improves the dynamic handling characteristics of the instrument[10]. The deflection and stiffness framework of Chapter 4 is directly applicable to minimising angular and lateral compliance at each joint, as positional error in PCMAs propagate and compound along the articulated arm [10]. Chapter 7 informs the treatment of shafts and shaft components the design of the rotating elements within each joint, specifically, the management of combined bending and torsional loads and the placement of stress concentrations away from critical cross-sections [10]. The rolling-contact bearing analysis of Chapter 11 is the main focus to joint pivot design, which covers load rating, fatigue life, and preload selection[8][10]. Nonpermanent joint analysis in Chapter 8 remains relevant for the bolted interfaces used in general PCMA segment assembly, where controlled preload ensures repeatable joint geometry across disassembly and reassembly cycles[10].

3 Research Methodology

The project focuses on the development and implementation of a joint design with position-holding technology on a PCMA that satisfies defined metrological and performance measurements. The project, therefore, uses a design research methodology that will aid in the design, testing, and building of the joint design.

3.1 Derivation of Design Requirements

The joint specifications will derived quantitatively from an existing error propagation model [7]. The error model relates joint run-out tolerances and encoder angular measurement to end-effector position ~~unreliability.~~ This will, in turn, establish measurable design requirements, including: maximum allowable run-out, holding torque, cable routing envelope, and encoder integration requirements. Data sheets, user and reference manuals from the three most popular PCMAs models used in industry will also contribute to the establishment of design targets, including:

- Faro Quantum X FaroArm series
- Hexagon (formerly Romer) Absolute Arm 85 series
- Kreon Onyx measuring Arm and Skyline scanner

This will also influence the modularity of the joint design.

[8]Bearing preload is an important lever for eliminating backlash and improving joint stiffness while minimising excessive friction [10]

3.2 Concept Development

Alternative joint mechanisms are generated through literature review and mathematical reasons from first-principles. Friction hinges and damping grease such as Nyogel would provide a baseline reference, despite their inability to meet all the design requirements. Concepts will be evaluated qualitatively against the derived requirements, with infeasible options disqualified earl~~ier~~ on. The shortlisted concepts are refined to a comparable level of detail before a preferred joint design is chosen and justified against the requirements.

3.3 Detail Design and Analysis

The chosen joint design will be developed in full, addressing all the aforementioned design requirements. Material selection dimensional tolerances will be informed by the run-out and compactness requirements and the sources obtained from documentation of the aforementioned PCMAs most popular in industry. Analytical methods including shaft deflection calculations, holding torque modelling [10], used alongside with Finite-Element Analysis (FEA) methods such as static structural analysis will be used to verify structural and mechanical behaviour [6]. The kinematic model of the joint will be formulated using a modified D-H model convention developed by Leon Hall [7].

3.4 Design Evaluation

The final design is evaluated against the established industriall~~y established~~ standards such as UNE-EN ISO 10360, ASME B89.4.1, and VDI/VDE 2617 [4], and where applicable, virtual simulation. The verification will confirm~~s~~ that individual joint requirements have been met. Validation will assess whether the joint contributes to overall arm measurement accuracy within acceptable limits. ~~Information obtained verification and validation and areas for further experimental validation will be acknowledged in the conclusion.~~

4 Workplan

The following activities are expected for the execution of the project. The costs and time scales associated with each activity is given in the following sections 5 and 7.

4.1 Conduct Literature Review and Industry Investigation

Review academic- and industry-facing sources on PCMA joint design, kinematic modelling, error sources, and calibration standards. Investigate the three most popular PCMAs in industry sing available data sheets, user manual, and reference manuals.

4.2 Derive Design Requirements

Using Hall's error propagation model with the extracted PCMA specifications, derive quantitative technical performance measures (TPM). These TPMs include maximum allowable run-out, minimum holding torque, cable routing envelop, encoder integration constraints, eight, and dimensional tolerances.

4.3 Develop and Choose Joint Concept

Generate joint concepts through first-principles reasoning and literature review. Evaluate concepts qualitatively and quantitatively against TPMs. This will eliminate infeasible options, and refine shortlisted designs to a comparable level of detail~~s~~ before justifying the ~~a~~ preferred concept.

Shaft deflection as in the arm lengths (stated in multiple papers)

4.4 Detail Design and Analysis

Develop the chosen concept in full. This will include bearing arrangement, holding torque mechanism, encoder integration, internal cable routing, and dimensional tolerances. Perform shaft deflection calculation, holding torque modelling, and FEA static structural analysis to verify mechanical behaviour. Formulate a modified D-H kinematic model incorporating force analysis due to the position-holding mechanism.

4.5 Building and Testing the Joint

Build a single joint prototype and conduct experimental testing, including SPAT to evaluate angular measurement accuracy, and holding torque measurements to verify position-holding performance. Verify result against the design requirements and established standards (UNE-EN ISO 10360, ASME B89.4.1, and VDI/VDE 2617), [4].

4.6 Evaluate Modularity Across PCMA Models

Assess whether the joint design meets the geometric and performance constraints of the three chosen PCMA models, which will confirm the modularity objective.

4.7 Document and Finalise Report

Consolidate all findings, analyses, and results into the final project report. Draw conclusions on whether the aim and objects were achieved, knowledge limitations, and propose recommendations for future work.

5 Resource Requirements

The successful execution of the PCMA joint design will require dedicated time, computational resources for CAD modelling as well as structural/kinematic analysis, and materials for manufacturing a prototype.The estimated costs have been divided into labour, materials and components, and computer/equipment usage, as seen in table 1 below[11, 12].

The labour is estimated at a rat of R450.00 per hour. The manufacturing phase includes the cost of raw materials and standard components needed to assemble the physical concept demonstrator. The testing phase includes costs associated with utilising calibration artifacts and metrology equipment to verify the joint's accuracy against the establishment design requirements.

Table 1: Estimated Cost per Activity

Activity	**Engineering Time**		**Running Costs**	**Facility Use**	**Capital Costs**	**PCMA Labour**		**PCMA Material**	**TOTAL**
	hr	**R**				**hr**	**R**	**R**	
Review Literature and Industry Investigation	25	11 250	500						11 750
Derive Design Requirements	30	13 500							13 500
Develop and Choose Joint Concept	40	18 000	100						18 100
Detail Design and Analysis	40	36 000	200					1 500	37 700
Building and Testing the Joint	100	27 000	500	2 000	500	120	8 500	1 000	93 500
Evaluate Modularity Across PCMA Models	40	18 000	500	500	500	15	500	2 500	29 250
Document and Finalise the Report	40	18 000	100	200					18 300
TOTAL	**315**	**141 750**	**1 900**	**2 700**	**1 000**	**125**	**8 500**	**5 000**	**222 100**

6 Expected Results

This project is expected to produce the following outcomes:

1. The validated joint design must meet all the quantitatively derived TPMs, including maximum run-out tolerances, minimum holding torque, and internal cable routing — without degrading the arm's ease of movement.

2. A modified D-H kinematic model that incorporates force analysis as a result of the position-holding mechanism. This model will provide a more complete framework for PCMA accuracy analysis and serve as a foundation for future arm development at Stellenbosch University.

3. SPAT verification results that confirmthe angular measurement accuracy of the joint falls within acceptable limits when compared against established standards (UNE-EN ISO 10360, ASME B89.4.1, VDI/VDE 2617).

4. A modularity assessment demonstrating that the joint design is geometrically and kinematically compatible with the three selected commercial PCMA platforms: the Faro Quantum X, Hexagon Absolute Arm 85, and Kreon Onyx.

5. A testing and calibration methodology for single-joint evaluation, designed to be replicable by future students or researchers continuing PCMA development at Stellenbosch University.

It is anticipated that the proposed position-holding mechanism will reduce arm drift to within the measurement uncertainty budget derived from Hall's error propagation model [7].

7 Project Schedule

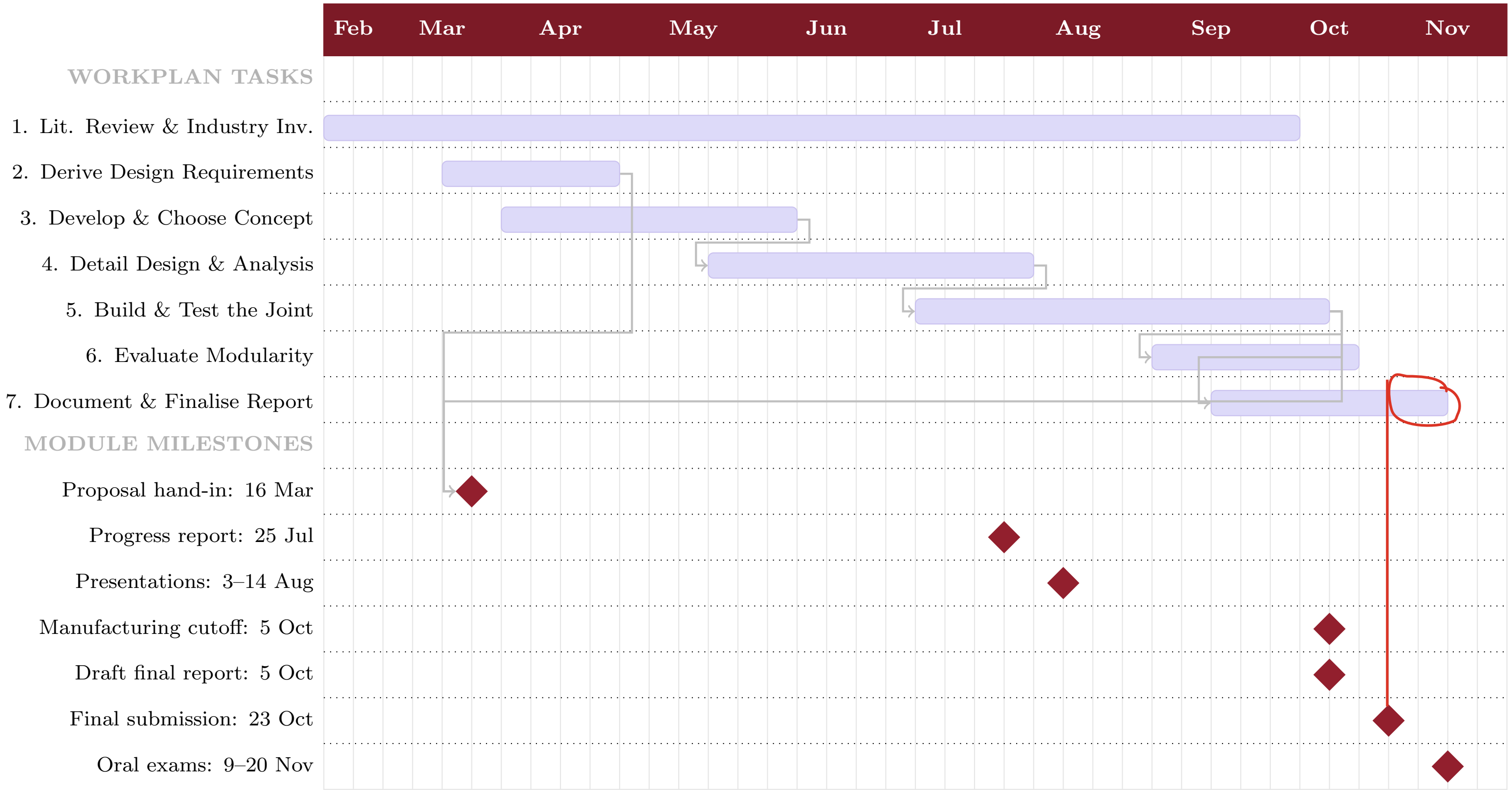

Figure 7: Project Schedule — M&M Skripsie 2026 (weeks from 9 February)

8 Risk Assessment

The following factors are used to conduct the risk assessment of the project:

- **Impact Scale:** Negligible (1), Low (2), Moderate (3), Significant (4), Catastrophic (5).
- **Likelihood Scale:** Very Unlikely (1), Not likely (2), Possible (3), Probable (4), Very Likely (5).
- **Risk Rating:** Low = [1–8], Moderate = [9–16], High = [17–25]. Impact × Likelihood = Risk Rating.

Table 2: Risk Assessment

Activity	Risk	Impact	Likelihood	Risk Rating	Mitigation
Detail Design & Analysis	Manufactured prototype does not meet derived run-out tolerances	4 (Significant)	3 (Possible)	12 — Moderate	Perform tolerance stack-up analysis before manufacturing; iterate design if required.
Build & Test the Joint	Position-holding mechanism introduces friction that impedes free arm movement	4 (Significant)	3 (Possible)	12 — Moderate	Analytically model holding torque vs. friction trade-off in detail design phase; prototype early.
Derive Design Requirements	Hall's error model is insufficient to capture f[illegible] effects of positio[illegible]lding mechanism	3 (Moderate)	3 (Possible)	9 — Moderate	Supplement model with Shigley's analytical framework; consult supervisor if gaps arise.
Build & Test the Joint	Workshop or lab access delays manufacturing and testing	3 (Moderate)	3 (Possible)	9 — Moderate	Book workshop time and equipment early; submit manufacturing request well before 5 Oct cutoff.
Derive Design Requirements	PCMA datasheets are insufficient or unavailable for modularity analysis	2 (Low)	3 (Possible)	6 — Low	Contact manufacturers and distributors directly; use published academic specifications as supplementary data.

Activity	Risk	Impact	Likelihood	Risk Rating	Mitigation
All Activities	Time overrun causes missed module deadlines	4 (Significant)	3 (Possible)	12 — Moderate	Adhere strictly to Gantt chart; flag delays to supervisor immediately; prioritise critical-path tasks.
Build & Test the Joint	SPAT results do not confirm accuracy within acceptable limits	4 (Significant)	2 (Not likely)	8 — Low	Revisit kinematic model and joint assembly; consult Ibaraki & Saito R-Test procedure for improved test setup [6].

9 Conclusions

This proposal presents a plan for the design, building, and testing of a joint for a portable coordinate measuring arm (PCMA) with position-holding mechanism. The project is motivated by the absence of commercially available joints that can satisfy the design requirements of kinematic accuracy, position-holding capability, internal cable routing, and compact lightweight form.

The engineering design methodology that will be followed begins with the quantitative derivation of joint specifications from Hall's error propagation model, followed by specifications from three widely used commercial PCMA platforms. These requirements will drive the generation and selection of the concepts, followed by detailed mechanical design and analysis using shaft deflection calculations, holding torque modeling, and FEA. A single physical prototype will be manufactured and tested against established metrology standards, with results used to verify and validate the design.

The project is expected to produce a validated joint design, a modified D-H kinematic model incorporating force analysis, and a replicable testing methodology — all of which will support future PCMA development at Stellenbosch University. Key risks have been identified and mitigation strategies defined.

The project schedule has been planned to meet all M&M Skripsie module deadlines, including the written proposal submission on 13 March, the progress report on 25 July, the manufacturing cutoff on 5 October, the final electronic submission on 23 October, and oral examinations in November 2026.

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A ECSA Strategy

A.1 GA1 - Problem-Solving

This project addresses a complex engineering problem, which is the absence of a joint design for PCMAs that satisfy kinematic accuracy, position-holding capability, internal cable routing, and compact and lightweight form. The problem is systematically defined in Section 1 through a review of existing limitations and the formulation of quantitative design requirements derived from Hall's error propagation model. Section 3 outlines the methodology used to generate and evaluate solution concepts using first principles, analytical models, and FEA. Section 6 defines the expected outcomes against which the solution will be validated, and Section 8 identifies and mitigates project risks.

A.2 GA2 - Application Scientific and Engineering Knowledge

The project applies knowledge from mechatronic engineering fundamentals throughout. In Section 3, shaft deflection calculations and holding torque modelling drawn from Shigley's Mechanical Engineering Design will be used to verify structural behaviour. The kinematic model of the joint is formulated using the Denavit-Hartenberg convention [7], extended with force analysis to account for the position-holding mechanism. FEA static structural analysis is applied to verify mechanical performance. Material selection is informed by stiffness-to-weight indices from engineering materials references.

A.3 GA3 - Engineering Design

The project follows a complete engineering design process as described in Section 3. Design requirements are derived quantitatively from an existing error propagation model, constraining run-out tolerances, holding torque, cable routing envelope, and encoder integration. Alternative joint concepts are generated through literature review and first-principles reasoning, evaluated against the derived requirements, and refined before a preferred concept is selected and justified. The chosen design is then developed in full detail, including bearing arrangement, holding torque mechanism, encoder integration, and dimensional tolerances.

A.4 GA5 - Use of Engineering Tools

Several engineering tools are applied critically in this project. CAD software is used for detailed design and dimensional analysis. FEA is used for static structural analysis to verify mechanical behaviour under loading. The Denavit-Hartenberg kinematic modelling framework, extended with force analysis, is used to derive and verify the joint contribution to end-effector accuracy. The Single Point Articulation Test is used as a standardised experimental tool to evaluate angular measurement accuracy. Each tool is selected based on its suitability for the task and its outputs are critically assessed against the design requirements and industry standards.

A.5 GA6 - Professional and Technical Communication

This report is structured to communicate the project clearly and professionally to a graduate engineering audience. The executive summary provides a concise overview of the project objectives, methodology, and expected outcomes. Each section follows a logical progression from background and motivation through to methodology, results, and conclusions. Figures, tables, and a Gantt chart are used to support and clarify the written content. References are cited in accordance with the IEEE standard. The report adheres to the department's technical writing guidelines, and AI tools have been used only for grammatical and formatting assistance, as declared in Section 1.5.

A.6 GA8 - Individual and Collaborative Teamwork

This project is executed independently, with regular supervision from Prof K. Schreve. The student is responsible for all aspects of the project, including literature review, derivation of design requirements, concept development, detailed design, manufacturing, testing, and reporting. The project plan, presented as a Gantt chart in Section 7, demonstrates structured time management aligned with the M&M Skripsie 2026 module deadlines. Progress will be reported to the supervisor at least fortnightly in the first semester and weekly in the second semester.

A.7 GA9 - Independent Learning Ability

The project requires the student to independently engage with and apply knowledge beyond the scope of taught modules. The literature review in Section 2 draws on academic journal articles, industry publications, and advanced textbooks to build a current understanding of PCMA joint design, kinematic modelling, and error analysis. Hall's error propagation model is studied and extended with force analysis to suit the specific requirements of this project. The student identifies gaps in existing literature and research, and applies this understanding to define a novel research direction. Source material is evaluated critically and used at the level appropriate to a final-year engineering project.